

HS Harz FuSa, Friedrichstraße 57

Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 14: Software Integration Testing

Version: 1.00

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1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	15:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit.
ADC	Analog to digital converter value read from an analog sensor input.
ASIL	Automotive Safety Integrity Level used by ISO 26262 for risk classification.
CAN	Controller Area Network, a vehicle communication bus.
ECU	Electronic Control Unit.
EUC	Equipment under Control.
FuSa	Functional Safety.
HW	Hardware.
QM	Quality Management level below ASIL.
SIL	Safety Integrity Level.
SW	Software.

4 Software Integration Testing

4.1 Sensor Boundary and Plausibility Tests

Tabelle 2 Sensor Boundary and Plausibility Tests

Test ID	Scenario	Input Values	Expected Output	Safety Parameter
SV-01	Normal driving baseline	ACC=300, GYRO=200, PRESSURE=250, Baby=OFF	D8 OFF, D9 OFF, D13 OFF, CAN normal	No unintended deployment
SV-02	Minimum acceleration	ACC=0, GYRO=200, PRESSURE=250	No deployment	Sensor range validation
SV-03	Maximum acceleration raw value	ACC=1023	Fault if outside valid range; lamp ON; air-bags OFF	Fault detection / safe state
SV-04	ACC just below threshold	ACC=699, GYRO=650, PRESSURE=250	No frontal deployment	Boundary testing
SV-05	ACC at threshold	ACC=700, GYRO=650, PRESSURE=250	Valid frontal crash; driver ON; passenger ON if Baby=OFF	Correct threshold behavior
SV-06	Pressure just below threshold	PRESSURE=649, ACC=300, GYRO=650	No side crash deployment	Boundary testing
SV-07	Pressure threshold at	PRESSURE=650, ACC=300, GYRO=650	Valid side crash; deployment allowed if no baby seat	Correct threshold behavior
SV-08	Gyro below safing threshold	ACC=800, GYRO=599, PRESSURE=250	No deployment if safing requires GYRO >= 600	Safing validation
SV-09	Gyro at safing threshold	ACC=800, GYRO=600, PRESSURE=250	Deployment allowed if no fault and no baby-seat suppression	Safing validation

4.2 Crash Function Validation Tests

Tabelle 3 Crash Function Validation Tests

Test ID	Scenario	Input Values	Expected Output	Safety Goal
CR-01	Valid frontal crash	ACC=800, GYRO=650, PRESSURE=300, Baby=OFF	Driver ON, Passenger ON, Lamp OFF, CAN crash status	SG2
CR-02	Valid side crash	ACC=300, GYRO=650, PRESSURE=800, Baby=OFF	Driver ON, Passenger ON or side output ON; CAN crash status	SG2/SG4
CR-03	One high sensor only	ACC=800, GYRO=300, PRESSURE=250	No deployment if safing condition not met	SG1
CR-04	Angular/rollover value only	ACC=300, GYRO=700, PRESSURE=250	No deployment without crash input	SG1
CR-05	Both crash sensors active	ACC=850, GYRO=700, PRESSURE=850, Baby=OFF	Deployment allowed; CAN crash status; no warning	SG2/SG4
CR-06	Crash followed by normal state	Crash values then normal values	Outputs reset according to prototype logic	State behavior

4.3 Passenger/Baby-Seat Suppression Tests

Tabelle 4 Passenger/Baby Seat Suppression Tests

Test ID	Scenario	Input Values	Expected Output	Safety Parameter
BS-01	Baby seat during frontal crash	ACC=800, GYRO=650, PRESSURE=300, Baby=ON	Driver ON, Passenger OFF, CAN passenger suppressed	Fail-safe suppression
BS-02	Baby seat during side crash	ACC=300, GYRO=650, PRESSURE=800, Baby=ON	Driver/allowed output ON, Passenger OFF	Priority logic
BS-03	No baby seat during crash	ACC=800, GYRO=650, Baby=OFF	Driver ON, Passenger ON	Correct normal deployment
BS-04	Baby seat, no crash	ACC=300, GYRO=200, PRESSURE=250, Baby=ON	Both airbags OFF	No unintended deployment
BS-05	Baby-seat input unstable/floating	Toggle Baby input or disconnect input	Fault or conservative passenger suppression; lamp ON if implemented	Input fault detection

4.4 Output Verification and Readback Tests

Tabelle 5 Output Verification and Readback Tests

Test ID	Scenario	Method	Expected Output	Status
OUT-01	Driver output command/readback	Command D8 HIGH during valid crash and read back actual output.	Readback equals command; otherwise, safe state.	Planned / extend code
OUT-02	Passenger output command/readback	Command D9 HIGH only if Baby=OFF and read back.	Readback equals command; otherwise, safe state.	Planned / extend code
OUT-03	Warning lamp output	Inject sensor fault and verify D13.	D13 HIGH during fault.	Partly implemented
OUT-04	Output stuck-at HIGH fault	Force/read output HIGH when command is LOW.	Fault detected; deployment inhibited; lamp ON.	Planned / test-board required
OUT-05	Output stuck-at LOW fault	Command output HIGH but readback LOW.	Fault detected; safe state or diagnostic fault.	Planned / test-board required

4.5 CAN Communication Tests

Tabelle 6 CAN Communication Tests

Test ID	Scenario	Input/Condition	Expected Behavior	CAN	Safety Parameter
CAN-01	Normal status frame	Normal sensor values	CAN ID 0x120, DLC 8, status normal, checksum correct		Communication integrity
CAN-02	Crash status frame	Valid crash scenario	Frame shows crash/deployment status and sensor values		Vehicle-level notification
CAN-03	Baby-seat suppression status	Valid crash + Baby=ON	Frame shows passenger airbag suppressed		Traceable safe action
CAN-04	Fault status frame	Invalid sensor or output mismatch	Frame shows fault code and warning status		Fault reporting
CAN-05	Wrong checksum received	Inject wrong checksum in received diagnostic/reset command	Command rejected; CAN error/fault logged		Data integrity
CAN-06	Missing CAN communication	Disconnect/suppress CAN receive if applicable	Fault or diagnostic warning according to requirement		Fault detection

4.6 Integration and System Validation Levels

Tabelle 7 Integration and System Validation Levels

What is Combined	Main Validation Question	Evidence
One software module at a time	Does each module meet its software requirement?	Module test report, coverage log.
Sensor manager + crash logic + deployment logic + fault handler	Does the software chain make the correct decision?	Serial logs, test script output.
Arduino/test board + sensors + outputs + CAN	Do real inputs produce correct real outputs?	Photos, videos, oscilloscope/logs, CAN logs.
Architecture vs SRS and safety concept	Does architecture include redundancy, diagnostics, safe state, CAN, backup power concept?	Review checklist.
Complete airbag ECU prototype	Does the ECU react according to safety goals and safety requirements?	Final validation report.

5 Figure and table listing

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6 List of dependent documents

No	Dokument name	Version	Date	File name
1	Airbag ECU Testing and Validation Plan	V3	31.05.2026	LCP13_Airbag_ECU_Testing_and_Validation_Plan_V3.pdf
2	Arduino Coding and Logic Implementation for Airbag ECU System	V1	02.06.2026	FuSa Report_Adruido.docx
3	Airbag ECU ASIL Code Division	V02	02.06.2026	Airbag_ECU_ASIL_Code_Division_V02.docx

7 Literature

- [1] International Organization for Standardization, ISO 26262-3:2018, Road vehicles - Functional safety - Part 3: Concept phase. Geneva: ISO, 2018. <https://www.iso.org/standard/68385.html>
- [2] International Organization for Standardization, ISO 26262-4:2018, Road vehicles - Functional safety - Part 4: Product development at the system level. Geneva: ISO, 2018. <https://www.iso.org/standard/68386.html>
- [3] International Organization for Standardization, ISO 26262-6:2018, Road vehicles - Functional safety - Part 6: Product development at the software level. Geneva: ISO, 2018. <https://www.iso.org/standard/68388.html>
- [4] International Organization for Standardization, ISO 26262-8:2018, Road vehicles - Functional safety - Part 8: Supporting processes. Geneva: ISO, 2018. <https://www.iso.org/standard/68390.html>
- [5] International Electrotechnical Commission, IEC 61508-1:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 1. Geneva: IEC, 2010.
- [6] International Electrotechnical Commission, IEC 61508-3:2010, Functional safety of electrical/electronic/programmable electronic safety-related systems - Part 3: Software requirements. Geneva: IEC, 2010.
- [7] ISO/IEC/IEEE, ISO/IEC/IEEE 29119-1:2022, Software and systems engineering - Software testing - Part 1: General concepts. Geneva: ISO, 2022.
- [8] International Organization for Standardization, ISO 11898-1:2024, Road vehicles - Controller area network (CAN) - Part 1: Data link layer and physical coding sublayer. Geneva: ISO, 2024.
- [9] AUTOSAR, Specification of RAM Test, AUTOSAR Classic Platform R24-11. AUTOSAR, 2024. <https://www.autosar.org>
- [10] MISRA Consortium, MISRA C:2025 - Guidelines for the use of the C language in critical systems. MISRA, 2025. <https://misra.org.uk>

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_14	14.06.2026	v1.00 Initial version. Software Integration Testing compiled from Testing Plan V3 Sections 8.1-8.5 and Section 10.